

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Local Search

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Local Decisions

Opening problem

A factory schedule has an enormous number of complete arrangements. Only the quality of the final schedule matters; the sequence of edits used to find it will be discarded.

**If every edit improves the schedule,
must the final schedule be globally best?**

Local Decisions

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A factory schedule has an enormous number of complete arrangements. Only the quality of the final schedule matters; the sequence of edits used to find it will be discarded.

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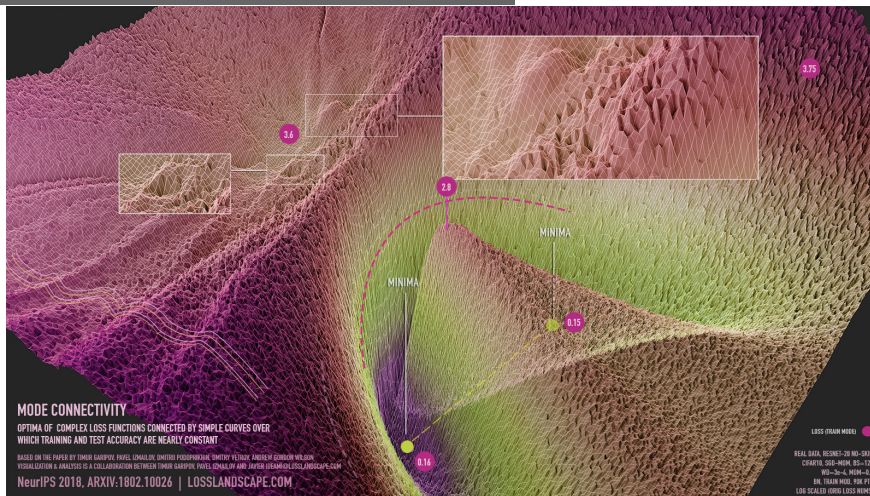
Memory question

If the path does not matter, why retain an entire frontier and reached table?

Escape question

If no neighboring state is better, should the search stop, restart, or temporarily get worse?

Why Global Search Is Difficult



A local method sees only a tiny neighborhood—not which distant basin is globally best.

A* and Local Search

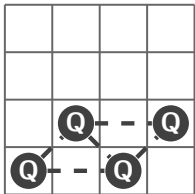
	A* graph search	Local search
State carried	Search node on a path	Complete candidate solution
Selection	Minimum frontier $g(n) + h(n)$	Neighboring state with a useful objective value
Memory	Frontier, best-cost records, parents	One state or a small population
Output	A path and its terminal state	A final state; path usually discarded
Strength	Systematic, cost guarantee under stated conditions	Scales to huge or infinite spaces
Risk	Frontier can dominate memory	Can miss entire regions or stop locally

Choose local search when the destination matters more than the route.

The Four-Queens Problem

Place four queens on a 4×4 board so that no two share a row, column, or diagonal.

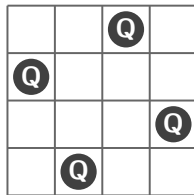
Conflicting placement



5 attacking pairs

Each dashed segment joins one pair.

Solution



0 attacking pairs

No two queens can attack each other.

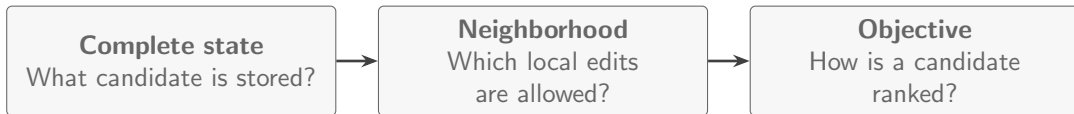
⇒
remove
conflicts

Try it live

Attacking pair = two queens sharing a row, column, or diagonal.

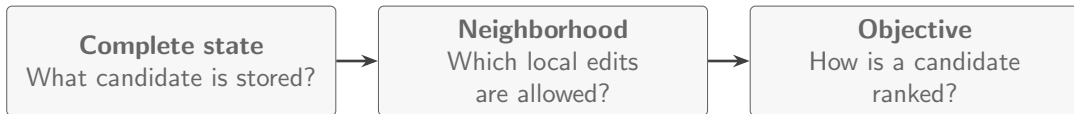
Three Design Choices

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize



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$h(s)$: conflict cost to **minimize**
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Representation

One queen in every column,
encoded by its row vector.

Move

Relocate one queen within its
own column.

Conflict cost

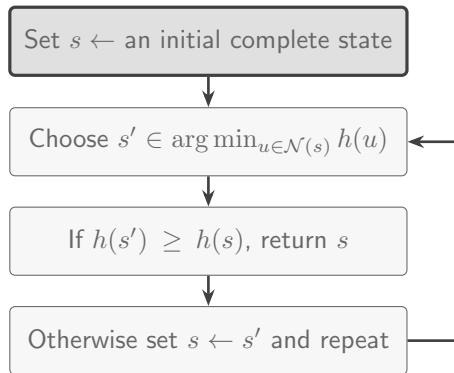
$h(s)$ counts pairs of queens
sharing a row or diagonal; a
solution has $h(s) = 0$.

Minimizing $h(s)$ and maximizing $V(s) = -h(s)$ induce exactly the same ordering.

Steepest Descent

$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize



One-step view

No lookahead: only the current state's neighbors affect the move.

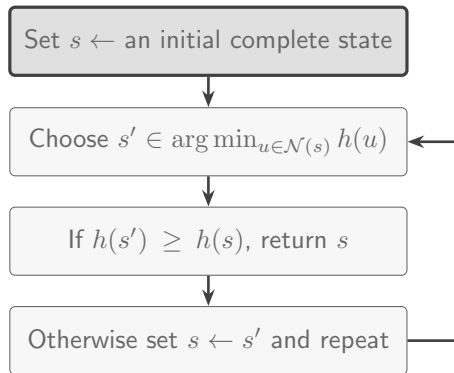
Small memory

Keep the current state and the best candidate seen while scanning neighbors.

Steepest Descent

$h(s)$: conflict cost to **minimize**

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No lookahead: only the current state's neighbors affect the move.

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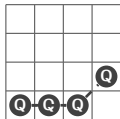
Keep the current state and the best candidate seen while scanning neighbors.

Stopping is local

No improving neighbor does *not* certify a global minimum unless the landscape has extra structure.

Four-Queens Model

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize



$[1, 1, 1, 2]$ $h = 4$

Representation

The c th vector entry is the row of the queen in column c .
One queen per column is built into every state.

Neighborhood

Move one queen to one of the other three rows:

$$|\mathcal{N}(s)| = 4(4 - 1) = 12.$$

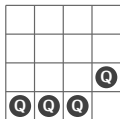
Cost

Count each unordered attacking pair once. Because $h(s) \geq 0$, reaching $h = 0$ certifies a solution.

Easy Four-Queens Trace

$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize



$[1, 1, 1, 2]$

$h = 4$

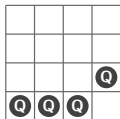
Choose the steepest improvement

Among the 12 neighbors, moving the column-3 queen from row 1 to row 4 gives the unique minimum cost, $h = 1$.

Easy Four-Queens Trace

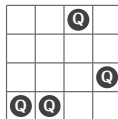
$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize



$[1, 1, 1, 2]$

$h = 4$



$[1, 1, 4, 2]$

$h = 1$

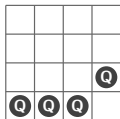
Improve again

From $[1, 1, 4, 2]$, moving the column-1 queen to row 3 removes the final attacking pair.

Easy Four-Queens Trace

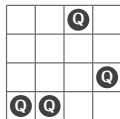
$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize



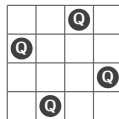
$[1, 1, 1, 2]$

$h = 4$



$[1, 1, 4, 2]$

$h = 1$



$[3, 1, 4, 2]$

$h = 0$

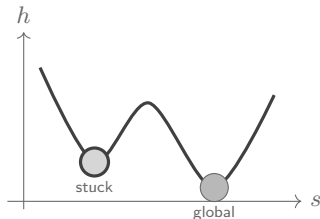
A certified finish

The trace reaches $[3, 1, 4, 2]$ with $h = 0$. Here the local stopping point is also globally optimal because no negative conflict count exists.

Landscape Traps

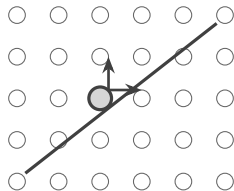
$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to **maximize**

Local minimum



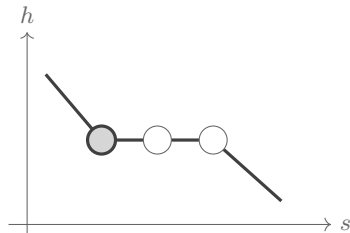
The current state beats every neighbor but not every state.

Ridge



The useful continuous direction lies between the available local moves.

Plateau or shoulder

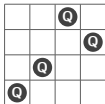


An equal-cost move may wander forever or reveal a later descent.

Basic hill climbing cannot distinguish a dead plateau from a useful shoulder.

Sideways Escape

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize



1243

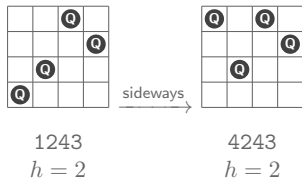
$h = 2$

Basic rule stops

State 1243 has no neighbor with cost below 2. Strict steepest descent returns a non-solution.

Sideways Escape

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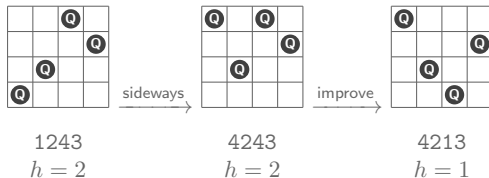


First shoulder

A bounded sideways move preserves $h = 2$ but changes which improving moves are available next.

Sideways Escape

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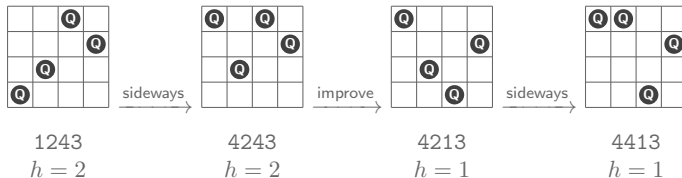


Progress, then another plateau

The search descends to $h = 1$, where strict descent again has no improving neighbor.

Sideways Escape

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize

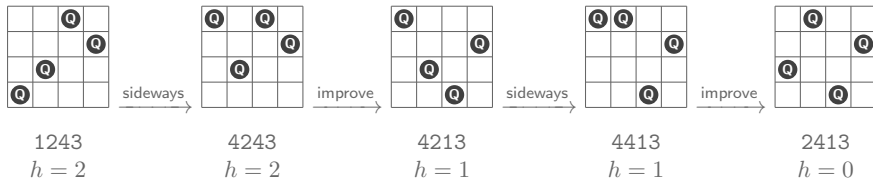


Second shoulder

Another equal-cost move opens a route to a solution. A cap prevents unbounded cycling among ties.

Sideways Escape

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Escaped

Two sideways moves expose two later improvements, ending at the solution 2413 with $h = 0$.

Hill-Climbing Variants

$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize

Variant	Move rule for minimizing h	Useful when	Main risk
Steepest descent	Evaluate neighbors; choose one with minimum h	Neighborhood is affordable to scan	Stops at local minima and plateaus
Bounded sideways	Permit $h(s') = h(s)$ up to a fixed cap	Plateaus often contain shoulders	Cycles or long wandering without a cap
Stochastic	Randomly choose among improving moves, biased by improvement	Multiple ascent/descent directions exist	Slower progress; still locally trapped
First choice	Sample neighbors until finding $h(s') < h(s)$	Thousands of successors make a full scan costly	May accept a weak improvement early

All four remain local: changing the move rule does not create a global frontier.

Random Restarts

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize



Independent success probability p

After r runs,

$$\Pr(\text{no success}) = (1 - p)^r.$$

Expected work

Let R include the final successful run:

$$\mathbb{E}[R] = 1 + (1 - p)\mathbb{E}[R] \implies \mathbb{E}[R] = \frac{1}{p}.$$

For the failures before it, $F = R - 1$:

$$\mathbb{E}[F] = \mathbb{E}[R] - 1 = \frac{1 - p}{p} \geq 0.$$

If $p > 0$, repeated independent restarts drive the failure probability toward zero.

Eight-Queens Evidence

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize

Empirical results from randomly generated 8-queens starts

Hill-climbing policy	Solved	Stuck	Steps if solved	Steps if failed
Strict steepest ascent/descent	14%	86%	4	3
At most 100 sideways moves	94%	6%	21	64

For total steps T , let L_s and L_f be the conditional mean steps for a successful and failed run:

$$\mathbb{E}[T] = pL_s + (1 - p)(L_f + \mathbb{E}[T]) \implies \mathbb{E}[T] = L_s + \frac{1 - p}{p}L_f.$$

Strict restarts: $p \approx 0.14$

$$\begin{aligned}\mathbb{E}[R] &= \frac{1}{0.14} \approx 7.14, \\ \mathbb{E}[T] &= 4 + \frac{0.86}{0.14}(3) \approx 22.4.\end{aligned}$$

Sideways restarts: $p \approx 0.94$

$$\begin{aligned}\mathbb{E}[R] &= \frac{1}{0.94} \approx 1.06, \\ \mathbb{E}[T] &= 21 + \frac{0.06}{0.94}(64) \approx 25.1.\end{aligned}$$

A higher success rate can still require longer individual runs.

Controlled Worsening

$h(s)$: conflict cost to **minimize**
 $V(s) = -h(s)$: equivalent value to maximize

Simulated-annealing rule

Sample one random neighbor s' and define

$$\Delta h = h(s') - h(s).$$

If $\Delta h \leq 0$, accept. If $\Delta h > 0$, accept with probability

$$\Pr(\text{accept}) = e^{-\Delta h/T}.$$

The same worsening move

Suppose $h(s) = 2$ and $h(s') = 4$, so $\Delta h = 2$.

T	$e^{-2/T}$	Interpretation
4	0.607	often escape
1	0.135	selective
0.25	0.00034	nearly frozen

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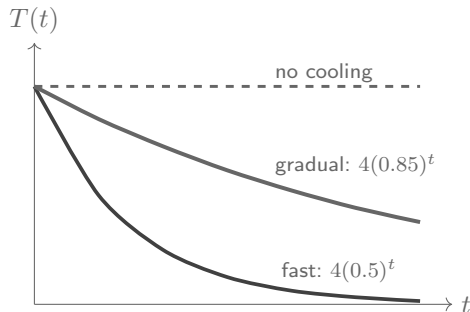
T	$e^{-2/T}$	Interpretation
4	0.607	often escape
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0.25	0.00034	nearly frozen

**Question: should a bad move be judged only by its cost,
or also by when it occurs?**

Cooling Schedules

$h(s)$: conflict cost to **minimize**

$V(s) = -h(s)$: equivalent value to maximize



Too fast

The process freezes before it can escape difficult basins.

Too slow

Exploration persists, but many accepted worsening moves delay convergence.

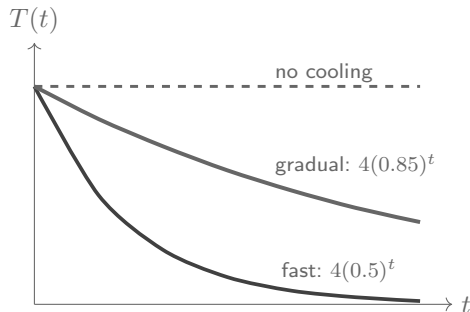
Schedule role

Temperature is an explicit budget for exploration: high early, selective later.

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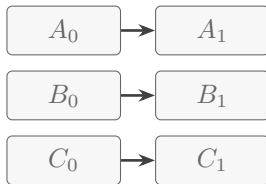
Temperature is an explicit budget for exploration: high early, selective later.

Sufficiently slow cooling gives an asymptotic global-optimum guarantee;
practical schedules trade that limit for speed.

Beam Search and Restarts

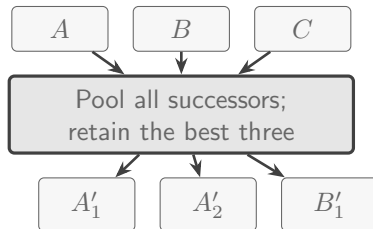
$h(s)$: conflict cost to **minimize**
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Three independent restarts



Each trajectory stays isolated; a promising run cannot redirect the others.

Local beam search with $k = 3$



The pooled choice shares information, but all k states can collapse into one region.

Stochastic beam selection trades some immediate quality for population diversity.

Genetic Algorithms: Selection, Crossover, Mutation

Eight-queens encoding

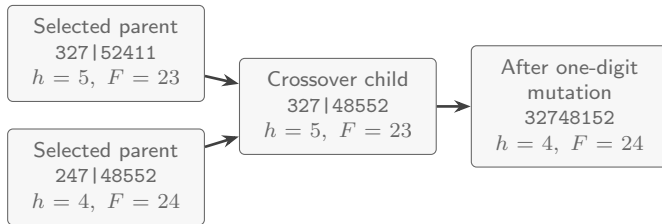
An 8-digit string records one queen's row in each column.

Fitness to maximize

There are $\binom{8}{2} = 28$ queen pairs.
Define

$$F(s) = 28 - h(s),$$

so higher fitness means fewer conflicts.



Two sources of exploration

Crossover recombines blocks from selected parents; mutation supplies local variation and can restore diversity.

Crossover is useful only when the representation preserves meaningful building blocks.

Method-Choice Challenge

Situation	Decisive requirement	Recommended starting point
Least-cost route and the path must be returned Any conflict-free queen placement; random starts are cheap	Systematic optimal path search Independent basin coverage	
Rugged objective; temporary worsening is acceptable Several promising regions should share evidence	Escape controlled by time Population with pooled successors	
Representation contains reusable partial structures	Recombination plus diversity	

Which method matches each requirement?

Method-Choice Challenge

Situation	Decisive requirement	Recommended starting point
Least-cost route and the path must be returned	Systematic optimal path search	A*, not local search
Any conflict-free queen placement; random starts are cheap	Independent basin coverage	Random-restart hill climbing
Rugged objective; temporary worsening is acceptable	Escape controlled by time	Simulated annealing
Several promising regions should share evidence	Population with pooled successors	Stochastic local beam search
Representation contains reusable partial structures	Recombination plus diversity	Genetic algorithm

Choose the escape mechanism and memory model that match the landscape.

Algorithm Comparison

Method	Memory	Escape	Use when	Condition or risk
A*	Frontier + records	Systematic expansion	Lowest-cost path matters	Optimal under A* conditions; high memory
Hill climbing	One state	None	Smooth landscape; cheap neighbors	Local optimum only
Random restart	One state per run	New independent basin	Random starts are cheap	Success probability $\rightarrow 1$ if $p > 0$
Simulated annealing	One state	Worsening moves via T	Rugged landscape	Schedule sensitive; slow-cooling theory
Local beam	k states	Pooled successors	Regions can share evidence	Population can lose diversity
Genetic algorithm	Population	Crossover + mutation	Representation has reusable blocks	Encoding and operator sensitive

No algorithm dominates across path requirements, landscape shape, and memory limits.

Challenge: Activate Both Modules

One complete candidate

$$x = \underbrace{x_1 x_2 x_3}_{\text{left module}} \mid \underbrace{x_4 x_5 x_6}_{\text{right module}} .$$

A bit is 1 when its component is active and 0 otherwise.

Local move

Flip exactly one bit. No path is retained; only the current design matters.

Before the reveal

Predict how hill climbing, random restarts, simulated annealing, local beam search, and a genetic algorithm behave on this same landscape.

A deceptive module value

If k components are active in one module, its value is

k	0	1	2	3
$q(k)$	3	2	1	4

Partial activation loses value; completing all three produces a jump.

Objective and goal

$$F(x) = q(k_L) + q(k_R) \quad (\text{maximize}),$$

with unique optimum 111|111 and $F = 8$.

Hill Climbing and Random Restarts

One steepest-ascent trace

Breaking the first tie by flipping the leftmost possible bit,

$$100|100_4 \longrightarrow 000|100_5 \longrightarrow 000|000_6.$$

The subscripts are F -values. Every neighbor of $000|000$ has $F = 5$.

Locally best, globally short

Strict hill climbing stops at $F = 6$, below the global value 8.

Uniform random restarts

For one three-bit module, steepest ascent reaches 111 exactly when the initial number of ones satisfies $k \geq 2$:

$$p_{\text{module}} = \frac{\binom{3}{2} + \binom{3}{3}}{2^3} = \frac{1}{2},$$

$$p = p_{\text{module}}^2 = \frac{1}{4}.$$

Both modules must begin in that basin. Therefore,

$$\mathbb{E}[R] = 4,$$

$$\Pr(\text{no optimum after } r \text{ runs}) = \left(\frac{3}{4}\right)^r.$$

Simulated Annealing

Acceptance rule for maximizing F

Improvements are accepted. For $\Delta F < 0$, accept with probability $e^{\Delta F/T}$. Use $T_t = 2(0.5)^t$ in this supplied trace from a one-module local peak.

Proposed move	ΔF	T	Acceptance test	Decision
$111 000_7 \rightarrow 111 100_6$	-1	2	$e^{-1/2} = 0.607 > \rho = 0.40$	accept
$111 100_6 \rightarrow 111 110_5$	-1	1	$e^{-1} = 0.368 > \rho = 0.20$	accept
$111 110_5 \rightarrow 111 111_8$	+3	0.5	improvement	accept

Activation valley

The first two moves deliberately lose value, crossing the partially active valley.

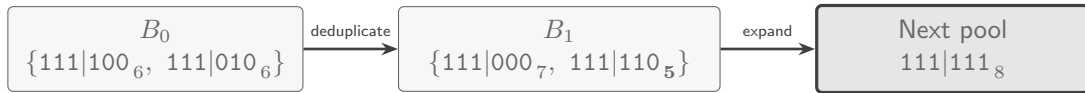
Cooling schedule

The same loss becomes less likely later: temperature budgets exploration early and selectivity later.

Local Beam Search

Reproducible local beam rule, $k = 2$

Generate successors only, remove duplicates, and retain the two highest- F states. In the relevant $F = 5$ tie, use the displayed bridge state $111|110$.



Deduplication opens a slot for the bridge

Both states generate the attractive backtrack $111|000_7$. After it occupies one slot, the lower-valued $111|110_5$ survives in the other and generates the optimum.

Beam search can retain a worse successor; it is not simply k independent hill climbers.

Genetic Algorithm

Selected parents and module-boundary crossover

111|000₇
000|111₇

crossover at the module boundary →

111|111₈
000|000₆

Why this crossover works

The encoding makes each complete module a reusable building block. The selected parents carry complementary high-value modules.

What it does not guarantee

Other parents or crossover points may produce worse children. Selection, initialization, and mutation still matter.

Challenge Debrief

Method	Challenge outcome	Decisive mechanism
Hill climbing	Stops at 000 000, $F = 6$	None; accept only improvement
Random restart	$p = 1/4$ per uniform start	Sample an independent basin
Simulated annealing	Supplied trace reaches $F = 8$	Accept temporary losses via T
Local beam, $k = 2$	Displayed beam reaches $F = 8$	Preserve a lower-valued bridge state
Genetic algorithm	Favorable crossover generates $F = 8$	Recombine complete modules

Interpretation

These outcomes are conditional on the displayed initial states and choices. None of the five methods guarantees the optimum within a finite budget.

Opening Questions Revisited

If every move improves the objective, must the final state be globally best?

Opening Questions Revisited

If every move improves the objective, must the final state be globally best?

No

Strict improvement can terminate at a local minimum, ridge, or plateau. The stopping rule sees neighbors, not the whole landscape.

Why discard the path?

When only the final configuration matters, retaining paths and a global frontier can spend memory on information the objective never uses.

How can search escape?

Use bounded ties, independent restarts, temperature-controlled worsening moves, or a diverse population.

Local search exchanges systematic coverage for scale.

Local-Search Takeaways

Formulate

Complete state, neighborhood, and objective determine the landscape.

Keep signs clear

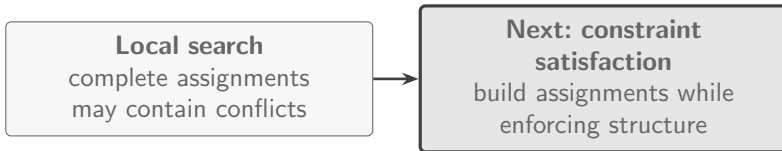
Minimize conflict cost h , or maximize a declared value such as $-h$ or $28 - h$.

Match the escape

Ties, restarts, temperature, and populations address different traps.

Measure

Track success rate, work per run, diversity, and schedule sensitivity.



Thank You

Any questions?

References

- T. Garipov, P. Izmailov, D. Podoprikin, D. Vetrov, and A. G. Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.